XVA calculator: validation report

EUR swaps, Hull-White 1F, Monte Carlo · Louis Delacour · November 2026 · version 0.0.1

> Self-review by the author. An independent review is still needed before any use beyond demonstration. All market data are synthetic or public (ECB); the counterparty is fictitious.

1. Summary

Check	Result	Verdict
Prices (IRS, OIS, swaptions, CDS) against QuantLib 1.44	errors < 1e-8 of notional; CDS < 0.1 bp	validated
EE* of a single swap against the closed-form swaption (Jamshidian)	max error 0.46 bp of notional, below the standard error (0.87 bp)	validated
Monte Carlo convergence	standard error in $N^{-0.504}$, antithetic gain ×4.1	validated
CRN sensitivities	sum of pillar deltas = parallel to within 1 €, SE < 1%	validated
Point-in-time backtest, directional positions	tails too fat at 1 month (KS p ≤ 0.005)	known limit
Point-in-time backtest, steepener	clear rejection at every horizon	out of scope
Sensitivity to unobservable parameters	vols ±20% → CVA ±16%; spreads ±25% → CVA ±23%	proxies to replace

The engine is fit for its intended use: CVA/DVA/FVA and pre-trade for a netting set of *directional* EUR swaps. It is not suited to a portfolio whose main risk is the slope of the curve.

2. Scope and model

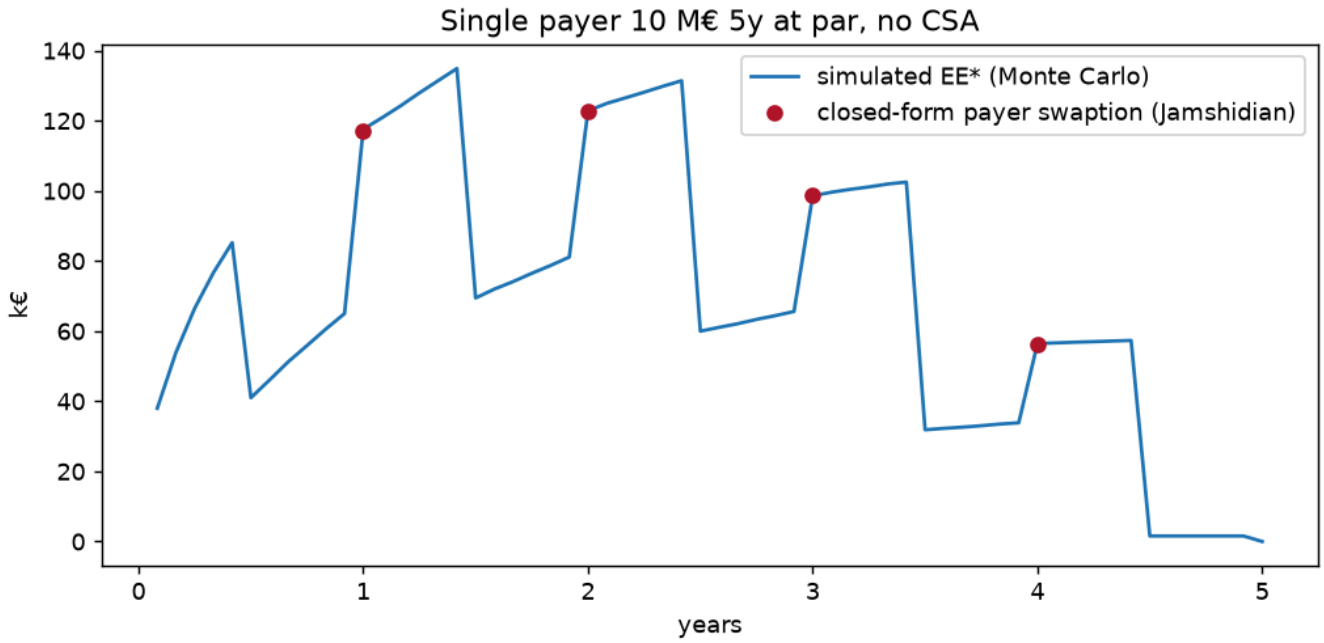
- **Products**: EURIBOR 6M IRS (deterministic 6M/€STR basis) and €STR OIS, payer or receiver, spot or forward start.
- **Model**: Hull-White 1F, exact joint simulation of (x, ∫x), unbiased discounting (E[D] = P(0, t)), antithetic variates. Q calibration: a = 0.03, σ fitted to the 10-year co-terminal ATM swaptions (σ = 0.84%).
- **Exposure**: monthly grid ∪ cash-flow dates ∪ t − MPoR nodes; netting; CSA with thresholds, MTA and MPoR; EE, PFE 97.5%, EEPE.
- **Credit**: piecewise-constant λ bootstrapped on synthetic CDS (BBB counterparty, A-rated bank), R = 40%.
- **Demo netting set**: 7 fictitious swaps, 435 M€ notional, 10,000 paths, 240 dates.

Demo netting set (k€, SE)	CVA	DVA	FCA	FBA
No CSA	659.7 (3.2)	227.8 (2.4)	81.2 (0.3)	34.2 (0.4)
CSA (threshold 10 M€, MTA 0.5 M€, MPoR 10 d)	446.3 (0.7)	159.5 (1.1)	57.0 (< 0.1)	23.5 (0.2)

The CSA removes 32% of the CVA. Pre-trade (ΔXVA of a new swap on the paths already simulated): 1 to 3 s.

3. Pricing and exposure tests

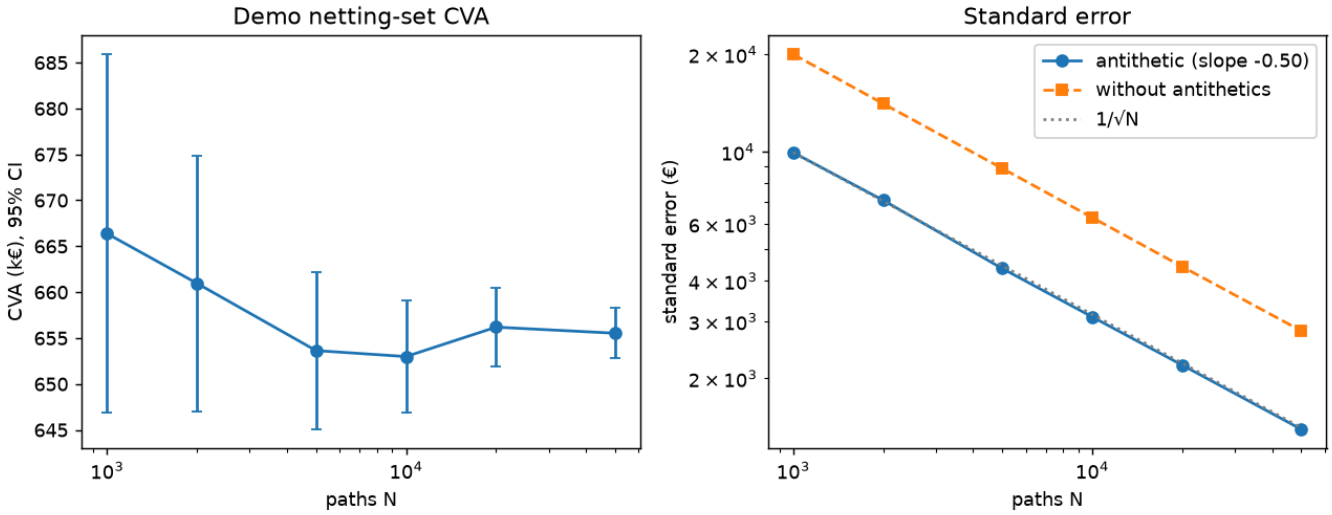
41 automated tests (pytest, about 45 s), including: martingale E[D] = P(0, t); ZC option and Jamshidian swaption against QuantLib; IRS and OIS against QuantLib; E[D·V(t)] equal to the value of remaining cash flows, including mid-period; PFE equal to the closed-form quantile within the binomial interval; sub-additive netting; perfect collateral at zero MPoR → 0; sum of Euler allocations = CVA; pre-trade = full recompute; bank CVA = counterparty DVA.



Without collateral, EE*(t) of a payer swap is the value of a payer swaption expiring at t on the residual swap. Max error 0.46 bp of notional with 40,000 paths (SE 0.87 bp). The saw-tooth follows the cash flows: floating received every six months, fixed paid every year.

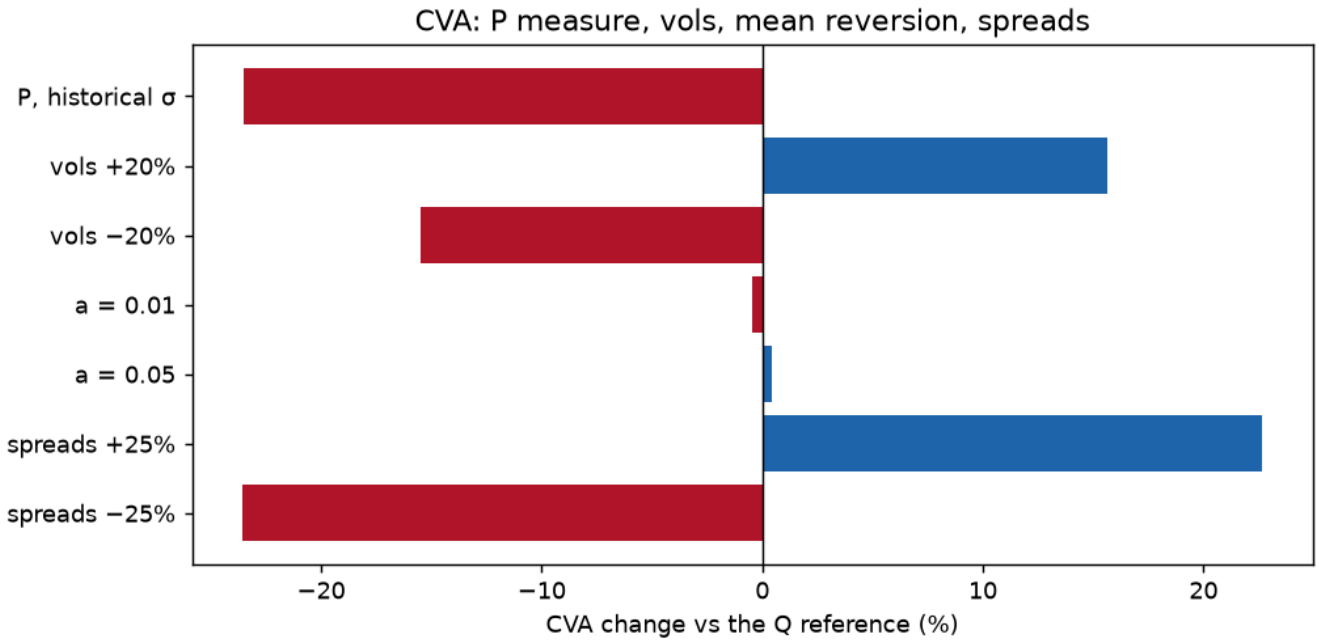
Error found and fixed: the formula for the running €STR leg given in the project sheet (exp(∫r)·P(t, e) − P(t, e)) is wrong; the correct value is exp($\int_s^t$ r) − P(t, e). The martingale test caught it.

4. Convergence



Demo netting set without CSA. 10,000 paths: CVA 653.0 k€ (SE 3.1), 1.4 s; 50,000: 655.5 k€ (SE 1.4), 6.6 s.

5. P measure against Q and unobservable parameters



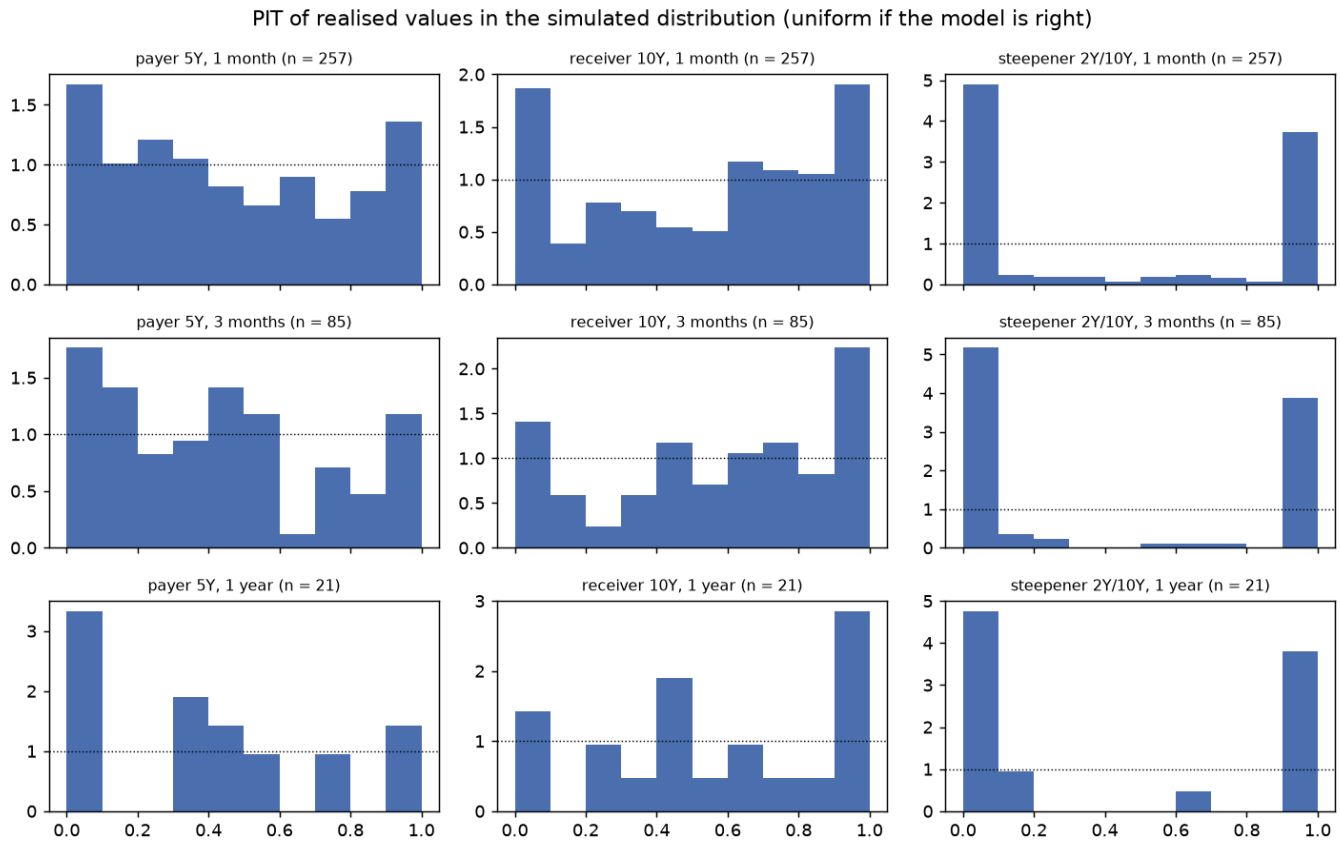
Same draws, 10,000 paths. Historical σ (ECB, one year): CVA −23.5%. Mean reversion weighs ±0.5% once σ is recalibrated.

A CVA computed under P would be a quarter lower: the implied vol premium weighs as much as a 25% shock to spreads. The two most influential inputs (swaption vols and CDS spreads) are currently synthetic.

6. Point-in-time backtest of exposures

Daily ECB AAA curves 2004-2026 (5,665 days). At each date, σ(P) is calibrated on the previous year (no look-ahead); the distribution of the value at the horizon is simulated from that day's curve and the realised value is ranked in it (PIT). Non-overlapping windows. Kolmogorov-Smirnov test of uniformity.

Horizon	Windows	KS power	Payer 5Y	Receiver 10Y	Steepener
1 month	257	55%	p 0.005	p < 0.001	p < 0.001
3 months	85	21%	p 0.02	p 0.03	p < 0.001
1 year	21	10%	p 0.08	p 0.14	p 0.001



Power computed against a true vol 25% higher. At 3 months and 1 year, power is too low to conclude.

Directional positions: PITs pile up at both ends (6 to 10% of realisations below the 2.5% quantile at 1 month). Historical tails are fatter than the Gaussian and σ(P) on a rolling year reacts late (2008, 2022). Steepener: with a single factor, 2-year and 10-year rates move together; the model sees almost no risk on a DV01-neutral position.

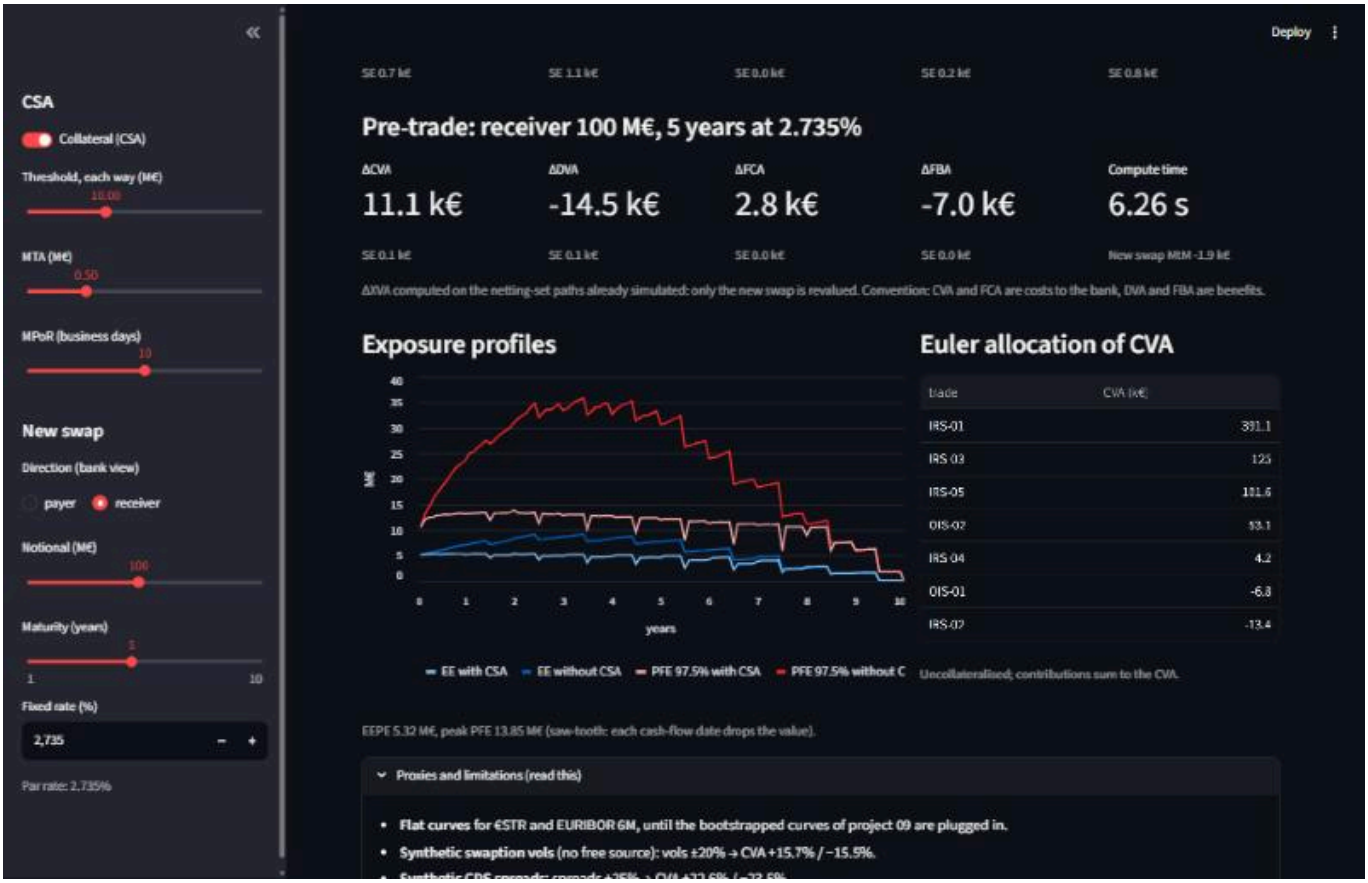
7. Limitations

Limitation	Measured or expected impact
Flat €STR / EURIBOR 6M curves (until the curves project is plugged in)	smoothed profiles; no initial slope
Synthetic swaption vols	±20% → CVA +15.7% / −15.5%
Synthetic CDS spreads	±25% → CVA +22.6% / −23.5%
One factor	slope risk invisible (backtest rejected)
Gaussian tails	PFE and EE understated in stress regimes
MTA added to the threshold, cash flows during the MPoR ignored	documented approximation; second-order effect on EE
No wrong-way risk, no seasoned trades, no TARGET2 schedule	not quantified
BA-CVA: risk weights copied from memory	to check against MAR50.16 before any regulatory use

8. Recommendations

1. Plug in the bootstrapped €STR and EURIBOR 6M curves and the TARGET2 dates (interface already in place).
2. Replace the synthetic vols and spreads with market data; while they stay synthetic, publish CVAs with the ranges from section 5.
3. Move to a two-factor model (G2++) before applying the engine to a portfolio exposed to the slope.
4. Calibrate σ(P) on a shorter or weighted window, or add a tail factor, for P-measure uses (limit PFE).
5. Add wrong-way risk (rates-credit correlation) and cash flows during the MPoR.
6. Have this report and the code reviewed by an independent third party.

Appendix: demo



Streamlit app: EE and PFE profiles with and without CSA, Euler allocation of CVA, proxies shown with their impact.

Reproduce: `pip install -e .[test,rapport]`, `pytest`, `python -m xva.rapport`, `streamlit run app.py`.